Bayesian Fraud Detection

From Prior Knowledge to Posterior Uncertainty

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Introduction to Bayesian Learning

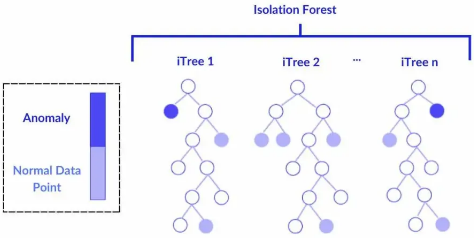
Context

- Modern financial systems generate high-volume transactional data with complex temporal patterns.
- Anomaly detection aims to identify rare or unusual transactions that deviate from normal behaviour.
- In real-world settings, true fraud labels are often unavailable or incomplete.
- This motivates the use of unsupervised methods and probabilistic models to estimate anomaly risk.
- Transaction-level anomaly detection is inherently a **rare event classification problem**, where class imbalance and uncertainty are central challenges.

Objective

Motivation

- Detect anomalous transactions without having true fraud labels.
- Use Isolation Forest to construct a proxy anomaly label.
- Show how Bayesian inference models uncertainty, not only point estimates.

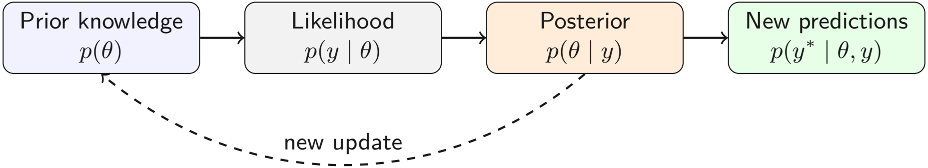


Bayesian Starting Point

Bayes' rule

$$p(\theta \mid y) = \frac{p(\theta)\, p(y \mid \theta)}{p(y)}$$

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{marginal likelihood}}$$



What Each Term Means

Components

- **Prior**: what we believe before observing the current data.
- **Likelihood**: how compatible the data are with each parameter value.
- **Posterior**: updated belief after observing the data.
- **Marginal likelihood**: normalization constant.

Marginal likelihood

$$p(y) = \int p(\tilde{\theta})\, p(y \mid \tilde{\theta})\, d\tilde{\theta}$$

Why the Marginal Matters

Conjugate models (from the course)

- **Beta–Binomial:** closed-form posterior for Bernoulli data.
- **Gaussian–Gaussian:** conjugate update for mean estimation with known variance.
- In both cases, the posterior is available analytically thanks to conjugacy.

General case

- In more realistic models, the integral is not available in closed form.
- This motivates simulation-based inference (MCMC) to obtain posterior distribution of θ

$$p(\theta \mid y) \propto p(\theta)\, p(y \mid \theta)$$

Workflow

Project presentation path

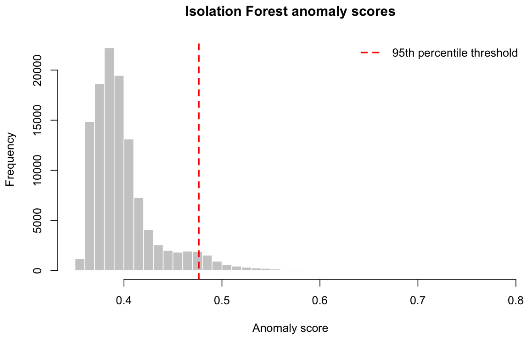
1. Data cleaning and feature engineering.
2. Build anomaly labels with Isolation Forest algorithm.
3. Start with a Beta-Binomial model for the global anomaly rate.
4. Track sequential posterior uncertainty over time.
5. Move to Bayesian logistic regression with transaction features.
6. Use MCMC for the posterior that cannot be obtained analytically.
7. Obtain metrics and uncertainty estimates for predictions on the test set.

Dataset Overview

- Anonymous bank transaction record dataset from 2017 to 2019.
- 116201 transactions in total.
- Dataset features: transaction date, withdrawal amount, deposit amount, cheque number, and account balance.
- Irrelevant variables (e.g. cheque number and textual descriptions) were removed and some features were engineered (e.g. log time, balance amount ratio).

Anomaly Label Construction

- True fraud labels are not available.
- Isolation Forest assigns an anomaly score.
- Transactions above the 95% quantile are labeled as anomalies.
- Therefore, the 5% anomaly rate is fixed by construction.



Code: Anomaly Label

Isolation Forest thresholding

```r
X <- data[, c("amount", "log_time", "balance",
              "is_withdrawal", "balance_amount_ratio",
              "weekday")]

model <- isolation.forest(X, seed = 123, nthreads = 1)
score <- predict(model, X)

threshold <- quantile(score, 0.95, na.rm = TRUE)
data$Y <- ifelse(score > threshold, 1, 0)
```

Beta-Binomial Model

Modeling precondition

- The 5% anomaly rate is a modeling precondition, as prior information.
- The first Bayesian model tests how this prior interacts with a random subsample.

$$Y_i \sim \text{Bernoulli}(\pi), \qquad \pi \sim \text{Beta}(\alpha_0, \beta_0)$$

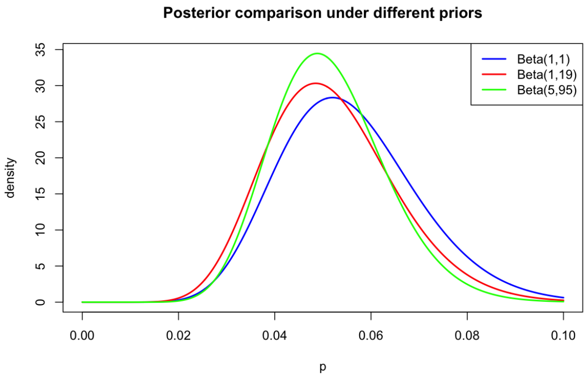
Closed-form update

$$\pi \mid Y \sim \text{Beta}(\alpha_0 + k,\ \beta_0 + n - k)$$

- k: number of observed anomalies; n: number of sampled transactions.
- Conjugacy gives a closed-form posterior.

Posterior Under Different Priors

- Random subsample: 250 transactions.
- Observed anomalies: 13.
- Informative priors pull the posterior toward 5%.
- Stronger priors reduce posterior uncertainty.



Sequential Bayesian Learning

Online update

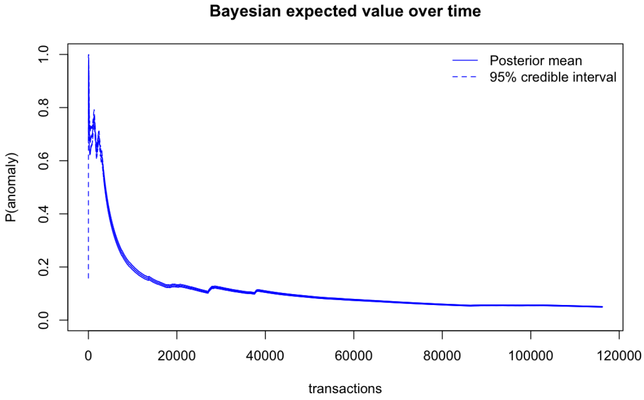
$$\alpha_t = \alpha_0 + k_t, \qquad \beta_t = \beta_0 + t - k_t$$

$$\mathbb{E}[\pi_t \mid Y_t] = \frac{\alpha_t}{\alpha_t + \beta_t}$$

- Here α_0, β_0 represent a non-informative prior.
- As more transactions are observed, uncertainty decreases.

Anomaly Rate Over Time

- Final posterior mean: $\mathbf{E}[\pi \mid Y] = 0.050007$.
- Final 95% credible interval for π variable: $[0.048960,\ 0.051063]$.
- The interval becomes very narrow with the full sequence.



Why Move Beyond Beta-Binomial?

Limitation of the global model

- Beta-Binomial estimates only global anomaly probability.
- It does not use transaction-level information, so no interpretability.
- Fraud risk should vary across transactions.
- Considering $Y_i \sim \text{Bernoulli}(\pi_i)$ with Y_i as anomaly indicator and π_i as the transaction anomaly probability:

$$\text{logit}(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}$$

Why MCMC?

Non-conjugate posterior

- Logistic regression has a non-conjugate likelihood.
- The posterior cannot be derived analytically.
- The marginal likelihood requires an intractable integral.

$$p(\boldsymbol{\beta} \mid Y, X) = \frac{p(\boldsymbol{\beta})p(Y \mid X, \boldsymbol{\beta})}{\int p(\boldsymbol{\beta})p(Y \mid X, \boldsymbol{\beta})d\boldsymbol{\beta}}$$

MCMC Intuition

Simulation from the posterior

$$\boldsymbol{\beta}^{(1)}, \boldsymbol{\beta}^{(2)}, \ldots, \boldsymbol{\beta}^{(S)} \sim p(\boldsymbol{\beta} \mid Y, X)$$

- MCMC samples plausible parameter configurations.
- Predictions average over many sampled models.
- This gives both posterior mean predictions and credible intervals.

Code: Bayesian Logistic Regression

MCMC fit with brms with Cauchy prior

```
bayes_formula <- Y ~ amount + balance + log_time +
  is_withdrawal + balance_amount_ratio + weekday

fit_bayes <- brm(
  bayes_formula,
  family = bernoulli(link = "logit"),
  data = train,
  prior = c(prior(cauchy(0, 2.5), class = "b"),
            prior(cauchy(0, 2.5), class = "Intercept")),
  chains = 4, iter = 1200, warmup = 600,
  seed = 123
)
```

Bayesian Logistic Regression Results with Cauchy Prior

Model specification

- Prior: Cauchy$(0, 2.5)$ (weakly informative)
- 92,960 training observations, 2,400 posterior draws

Key feature effects (log-odds scale)

- **balance amount ratio (-22.26):** strongest effect; high imbalance strongly indicates anomalies.
- **log time (-1.84):** later transactions are less likely to be anomalous.
- **Sunday (+5.57):** extremely high anomaly concentration on Sundays.
- **is_withdrawal (+0.34):** withdrawals slightly more suspicious than deposits.

MCMC Diagnostics and Convergence

Prior	Max $\hat{R}$	Min bulk ESS	Min tail ESS
Normal(0,1)	1.003	1045.7	1251
Normal(0,2)	1.005	831.1	1318
Cauchy(0,2.5)	1.003	807.8	1150

- $\hat{R} \approx 1$ indicates that the different chains are mixing well and converge to the same posterior distribution.
- Effective sample sizes are sufficient for fixed effects.

Classification Metrics

Confusion matrix quantities

$$\begin{aligned} TP &= \text{true positives} & TN &= \text{true negatives} \\ FP &= \text{false positives} & FN &= \text{false negatives} \end{aligned}$$

Metric definitions

$$\text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

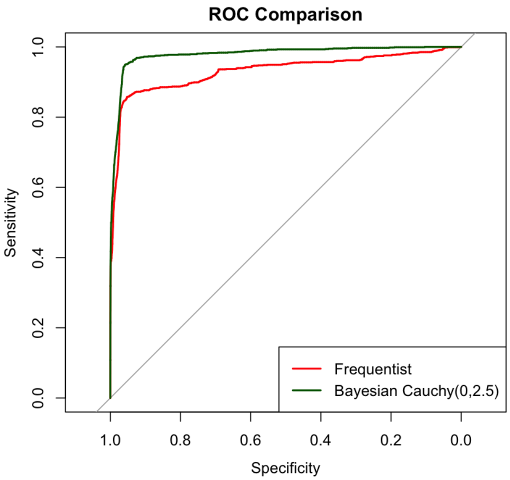
Predictive Performance

Model	Threshold	AUC	Recall	Precision	F1
Frequentist	0.50	0.9313	0.4707	0.7822	0.5877
Bayes N(0,1)	0.50	0.9767	0.6103	0.7919	0.6894
Bayes N(0,2)	0.50	0.9781	0.6164	0.7874	0.6915
Bayes Cauchy	0.50	0.9790	0.6198	0.7849	0.6927
Cauchy (opt.)	0.1493	0.9790	0.9172	0.5872	0.7160

Key insights

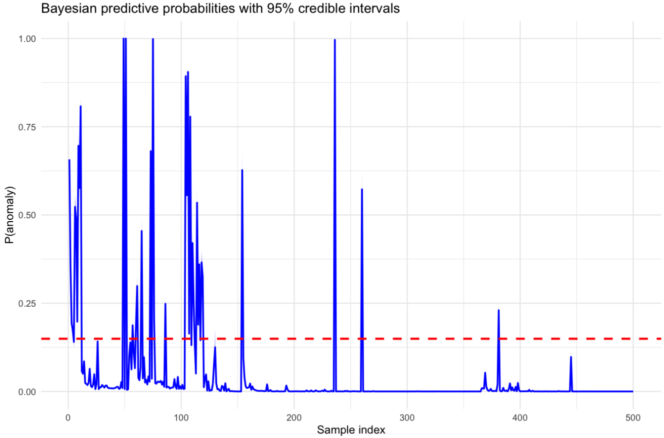
- Bayesian models consistently improve metrics performance.
- Lower optimal threshold significantly increases recall, reducing missed anomalies (false negatives) that would otherwise go undetected.
- This improvement comes at the cost of lower precision due to more false positives.

ROC Comparison



Posterior Predictive Uncertainty

- Each prediction is a distribution.
- Credible intervals flag uncertain cases.
- This is useful for manual review and risk scoring.



Takeaways & Limitations

Summary

- Bayesian inference turns assumptions and data into posterior distributions.
- MCMC is needed when the marginal integral is not analytically available.
- Bayesian logistic regression gives competitive predictions and uncertainty.

Main limits

- The target is an Isolation Forest proxy label, not confirmed fraud.
- Performance measures consistency with the labeling rule.
- A production setting would require temporal validation and true fraud outcomes.

References

Liu, F. T., Ting, K. M., and Zhou, Z.-H. (2008). Isolation Forest.

Gelman, A., Jakulin, A., Pittau, M. G., and Su, Y.-S. (2008). A weakly informative default prior distribution for logistic and other regression models.

Watsky, A. Extracted bank account statements of various bank accounts.

Repository: github.com/giolagioia/Bayes-Anomaly-Detection